American Board of Family Medicine

2024 IN–TRAINING EXAMINATION

SCORE RESULTS HANDBOOK

2024 In-Training Examination Score Results Handbook

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American Board of Family Medicine, Inc. (ABFM®)

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December 2024

Dear Program Director:

Thank you very much for participating in this year's In-Training Examination (ITE). As you know, the ITE provides the opportunity for you to gauge the academic progress of the residents in your program. It also provides residents with an opportunity to simulate the experience of taking the certification examination and given the psychometric comparability to the Family Medicine Certification, get an estimate of the likelihood of passing the exam via the Bayesian Score Predictor on RTM.

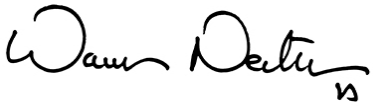
The new ABFM blueprint has been applied to this year's ITE, emphasizing the types of care—Acute, Emergent and Urgent, Chronic, and Preventive Care—in addition to the Foundations of Care. We believe that those who take the examination will not notice significant changes in the exam itself. However, the score reports will be enhanced to provide more useful feedback. We will be getting together program directors to give us feedback in January. Please note that you will be able to access your program's results via the RTM.

This handbook is intended to provide you with statistical analyses and explanations that will help you assess how the residents in your program performed and allow comparison to other programs. We appreciate that you have always utilized this information in the way that it has been intended to be used. If you need assistance regarding your results or how to access them, please do not hesitate to contact our Support Center at 877-223-7437 or Kathy Botner at 859-429-7360. If you have questions about the interpretation of the results, please contact our Director of Psychometrics and Innovation, Dr. Ting Wang at twang@theabfm.org. Do not hesitate to contact us if you should have any questions about the material in this handbook.

We look forward to working closely with you in our continuing efforts to improve all of our certification activities. At the Residency Leadership Summit (formerly PDW), we will have a workshop on using our tools to help guide development of residents. We have also created a toolkit of resources to help you and your residents in understanding and navigating ABFM Family Medicine Certification—and look forward to getting your feedback at about how that can be more useful. We look forward to talking with you.

Thank you for all you do.

Sincerely yours,



Warren Newton, M.D., MPH
President and Chief Executive Officer

TABLE OF CONTENTS

Introduction 7
In-Training Exam Summary Statistics 8
Table 1. Summary Statistics for the 2024 ITE 8
Table 2. Residents' Scaled Scores on the ITE 2024 by Year of Residency 8

Section I: Resident-level Reports
Overview 9
In-Training Examination Score Report 9
Header 9
Mean Scaled Score for PGY_ 9
Scaled Score 9
Score Report Body 10
Content Area 10
Exam Blueprint Percentage 10
Scaled Score 10
Standard Error 10
Bar Chart 10
Figure 1a. Resident's Score Report (Blueprint Category) 11
Figure 1b. Resident's Score Report (Body System Category) 11
Resident's Item Performance Report 12
Figure 2a. Item Performance Report (Blueprint Category) 13
Figure 2b. Item Performance Report (Body System Category) 13
Suggestions for Reviewing Scores with Residents 13

Section II: Program-level Reports
Overall Scaled Score Distribution Graphs 14
Mean Scaled Score Graph 14
Eight Year Trend Chart and Data 14
Figure 3. Scaled Score Distribution (PGY1-PGY4 and Total) 15
Figure 4. Example of a Program-Level Report 16
Figure 5. Data File for Trend Chart 16
Figure 6. Trend Chart for Residency Programs 17

Section III: Individual-Level Score Report Data files for All Residents
Introduction 18
Row Descriptions 18
Column Descriptions 18
Figure 7. Individual-Level Score Report Data Table for All Residents 19
Figure 8. Labels for the Data Table 19

Section IV: Instructions for Scoring by Hand
Introduction 20
Key Changes and Questions Removed from Scoring 20
Scoring by Hand 20
Table 3. Converting Raw Scores to Scaled Scores 21

Table 4. Questions by Blueprint Category 22

Section V: Bayesian Score Predictor
Overview 24
Using the Bayesian Score Predictor 24
Figure 9a. Bayesian Score Predictor in its Initial State 26
Figure 9b. Bayesian Score Predictor with PGY2=270… 26
Figure 9c. Bayesian Score Predictor with PGY2=410… 27

Section VI: Psychometric Summary
Purpose 28
Criterion-related Validity 28
Test Specifications 28
Equating the ITE to the FMCE 29
Scoring 29
2024 Administration 29
Reliability and Person/Item Fit 29
Table 5. Summary of 2024 ITE Candidates 31
Table 6. Summary of 2024 ITE Items 31

Introduction

This handbook provides information on how to interpret the reports associated with this year's *In-Training Examination* (ITE). These reports provide performance information about individual residents, the residency programs, and the nation. We would like to thank the Association of Family Medicine Residency Directors (AFMRD) and all the program directors whose substantive input guided the development of these reports. There are six sections to this document.

1) Resident-level reports and their interpretation.
2) Program-level reports and their interpretation.
3) Data files for both individual and program level data.
4) Instructions for manual scoring.
5) Bayesian Score Predictor use and interpretation.
6) Psychometric summary

SECTION I focuses on the interpretation of the individual resident's reports. Examples of the individual resident's *Examination Performance Report* and *Item Performance Reports* are provided, along with an explanation of each area of the report.

SECTION II provides examples of program-level reports. The *Mean Scaled Score* bar graph shows the program's average score by training year alongside the national average for that training year. The *8-Year Trend Chart* also provides program-to-national score comparisons for each program year of residency and the overall performance for the last eight years.

SECTION III provides resident-level subtest information (data files). Please note that these subtests are on the same scale (200-800) as the overall scaled score.

SECTION IV provides instructions for scoring the examination manually. It includes a list of any questions that were deleted from scoring, a raw score to scaled score conversion table, and a roster of which content category each question represented.

SECTION V explains what the Bayesian Score Predictor is and how to use it. The Bayesian Score Predictor can be found on the ABFM's website (https://rtm.theabfm.org/bayesian/predictor).

SECTION VI provides some summary information regarding the psychometric performance of the ITE on this administration. This could be useful to researchers.

If you have any questions, please contact us:

General questions	Support Center	**(877) 223-7437**	help@theabfm.org
Using the RTM System	Kathy Botner	**(859) 429-7360**	kbotner@theabfm.org
Scoring / Bayesian Score Predictor	Ting Wang, Ph.D.	**(859) 554-2093**	twang@theabfm.org
Bayesian Score Predictor	Thomas O'Neill, Ph.D.	**(859) 687-2538**	toneill@theabfm.org

In-Training Exam Summary Statistics

Table 1 provides summary statistics for the 2024 ITE. The statistics include the number of Accreditation Council on Graduate Medical Education (ACGME) accredited residency programs participating in the ITE, as well as the average size of the residency program and the associated standard deviation. Results from international residency programs are not included in this document.

Table 1. Summary Statistics for the 2024 ITE

Total Number of Programs Participating:	741
Mean Participating Residency Size:	20.2
Standard Deviation of Residency Size:	9.9

Table 2 provides a breakdown of resident's reported scaled scores by year of residency. As expected, the scaled score values tend to increase as residents advance through their program. The averages for PGY1 to PGY3 ranged from 414 to 494, a difference of 80 scaled score points. Program year 1 to program year 2 exhibits the greatest increase in scores, as indicated by an improvement of 48 scaled score points. Please note that PGY4 includes only residents who are in their 4th year of residency and are participating in the Length of Training (LoT) pilot study. PGY4 does not refer to residents who are in a dual specialty program (family medicine & psychiatry, family medicine & internal medicine, etc.)

Table 2. Residents' Reported Scaled Scores on the ITE 2024 by Year of Residency.

Year of Residency	Total Test	N	Mean	Standard Deviation
1	Scaled Score	5,279	414.1	77.8
	Percent Correct	5,279	57.8	6.5
	Raw Score	5,279	112.1	12.5
2	Scaled Score	4,957	462.2	76.3
	Percent Correct	4,957	61.7	6.1
	Raw Score	4,957	119.8	11.9
3	Scaled Score	4,727	494.0	77.5
	Percent Correct	4,727	64.3	6.1
	Raw Score	4,727	124.7	11.8
4	Scaled Score	35	554.6	68.4
	Percent Correct	35	69.0	5.1
	Raw Score	35	133.8	10.0
Total	Scaled Score	14,998	455.5	84.0
	Percent Correct	14,998	61.1	6.8
	Raw Score	14,998	118.6	13.2

SECTION I

Resident-level Reports

Overview

For each resident that takes the ITE, the residency program will receive two resident-level reports related to the resident's individual performance. Residents can download their own reports by accessing their physician portfolio on the ABFM website. The first report, the *2024 In-Training Examination Performance Report* (Figures 1a & 1b), presents overall performance and performance within specific content areas. This report closely resembles the format of the score reports used in the Family Medicine Certification Examination (FMCE). This is intended to familiarize residents with the type of feedback they will receive following their board certification examination. The first page of this three-page report explains how to interpret the results, while the next two pages represent the resident's performance broken out by the blueprint categories and by body system categories.

The second resident-level report, the Resident's 2024 Item Performance Report (Figures 2a and 2b), organizes the questions by both blueprint and body system, and then indicates whether each question was answered correctly or incorrectly. Those questions answered incorrectly by the resident are shown with bold red font, whereas those answered correctly are shown with regular green font.

In-Training Examination Score Report

Header. Please refer to the sample report in Figures 1a & 1b. The score report provides resident-level performance information on the 2024 ITE. At the top of the report, you will see (1) the resident's ABFM ID# along with (2) their name. You will also see the (3) Mean Scaled Score for Year 1 [or Year 2 or Year 3 as appropriate], (4) Scaled Score, and the (5) Number of Unanswered Items.

Mean Scaled Score for PGY_. This represents the mean scaled score for those residents who are in the same year of residency (1, 2, or 3) as the resident who is receiving the report. Comparing an individual resident's score to the appropriate mean can provide a gross indicator of how the candidate performed relative to other residents with comparable amounts of training.

Scaled Score. This represents an estimate of how the resident would perform on the FMCE if they had taken it instead of the ITE. Presently, the minimum passing standard on the certification examination is 380. If the resident's score on the ITE was just barely 380, they would be expected, theoretically, to have only a 50-50 chance of passing the certification examination. A score of 440 would increase the probability of passing to over 90%.

Of course, it is important to consider that there are going to be other intervening factors that can affect the outcome on the certification examination as well. For example, a resident's score on the certification examination could increase due to additional study and training or the score could decrease if the resident reduced the amount of study time in the months before the examination. Similarly, stressful events (marriage, divorce, serious illness, death of a loved one, etc.) could impede a resident's ability to study in a focused manner and subsequently have a negative impact.

Although the intent is to connect the results from the ITE to the FMCE, it is important to keep in mind that the ITE is not a replacement for the FMCE. The certification examination enjoys a much higher level of security in its administration, has more rigorous item review processes, and has a higher degree of precision due to the larger number of questions administered and the targeting of those questions. To place this in appropriate perspective, appreciate that the fees charged for each examination are reflective of ABFM's relative costs for developing each, approximately $65 for the ITE versus $1300 for the FMCE.

Score Report Body

The lower portion of each score report has four columns and a bar chart. Both the *Blueprint Category* chart (Figure 1a) and the *Body System Category* chart (Figure 1b) have the same columns and similar bar charts. The only difference is that a different content category structure is used to classify the questions.

Content Area. The first chart uses the Blueprint Category structure to organize the results. The second chart organizes the results by Body System Category. Although a question may be theoretically related to more than one content area, each question is categorized as being in only one content area within a chart.

Exam Blueprint Percentage. The exam blueprint percentage is the percentage of questions that are specified by the ITE and FMCE's test plan specifications. This information might be useful for deciding what content areas to focus on when preparing for the certification exam. The second chart (Figure 1b), which displays results by body system category, does not indicate the percentage of questions that will appear on the exam by each body system category. This omission is due to the examination being assembled exclusively according to the blueprint rather than by body system category. Also, be aware that that the percentage of questions in each category on a particular version of the ITE might be off by a percentage point or two.

Scaled Score. The scaled score is a transformation of the number of correctly answered questions that adjusts for the difficulty of the test in such a way that the results of different versions of the examination have the same meaning. On the ITE and FMCE, scaled scores can range from 200 to 800. Scaled scores in specific content areas are based upon responses to questions on that topic only. The "Overall Performance" scaled score is based upon responses to all the scorable questions on the examination.

Standard Error. The standard error (SE) can be understood as being an indicator of the "thickness" of a scaled score. It is helpful to think of all scaled scores, both overall performance and in specific content areas, as having a thickness rather than as being a point. The SE can be used to create a confidence interval around the scaled score. Plus-or-minus 1 SE creates a 68% confidence interval around a scaled score, meaning that it is expected that the "true ability" scaled score will fall somewhere in that range 68% of the time. Plus-or-minus 2 (actually 1.96) SEs creates a 95% confidence interval. When content areas have only a few questions, the SE is larger, but even the scaled score for the entire examination has an SE appreciably greater than zero. Nevertheless, it is still sufficiently small to make a score precise enough to be useful.

Bar Chart. The bar chart on the right side of the report depicts the data from the table on the left. For each content area, the resident's performance is represented by a bar. The midpoint marker of the bar represents their scaled score. The entire bar represents the 68% confidence interval (± 1 SE) around their scaled score. When reviewing a resident's performance by content area, please consider the precision of the

scaled score. Also, be advised that the ABFM only reports scores between 200 and 800. If a resident's score is below 200 or above 800, it will be reported as 200 or 800, respectively.

Figure 1a. Resident's Performance Report (Blueprint Category)

Score Description	Exam Blueprint Percentage	Scaled Score	Standard Error
Performance By Blueprint Category			
Acute Care and Diagnosis	35%	530	±69
Chronic Care Management	25%	590	±82
Emergent and Urgent Care	20%	630	±97
Preventive Care	15%	690	±109
Foundations of Care	5%	490	±187
Overall Performance	100%	590	±41

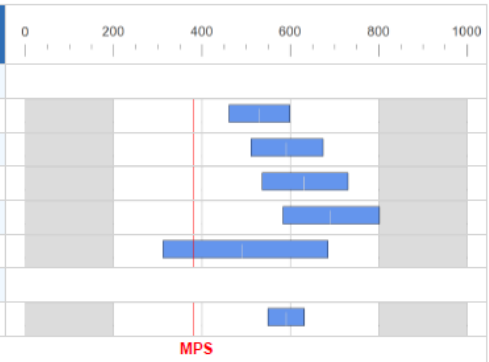
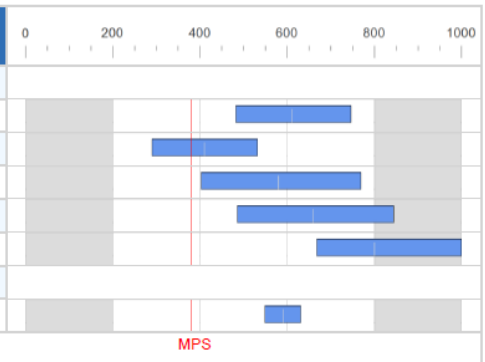


Figure 1b. Resident's Performance Report (Body System Category)

Score Description	Scaled Score	Standard Error
Performance By Body System		
Cardiovascular	610	±132
Injuries/Musculoskeletal	410	±121
Respiratory	580	±183
Psychiatric/Behavioral	660	±179
Sexual and Reproductive	800	±190
Overall Performance	590	±41



Resident's Item Performance Report

The Resident's Item Performance Report is also presented across Blueprint Categories (Figure 2a) and across Body System Categories (Figure 2b). The questions are identified using their sequence number from the test. Those questions answered incorrectly by the resident are shown in bold, red font. Those questions answered correctly are shown in regular, green font. This report can be used to help residents understand the types of questions typical in each of the content categories and to identify content helpful in preparing for the certification examination. Additionally, candidates may use the questions in bold red to identify content areas in which they could gain the most from studying.

Please note that questions marked with a "P" following the question number (e.g., 100P) indicate that the question was deleted from scoring for psychometric reasons and should not be used when scoring the ITE by hand using Table 3. However, even though psychometrically deleted questions did not function as expected, reviewing them may still have some utility for an individual resident and so they have been included on these reports. When a question is deleted from the examination for content reasons, such as when there were multiple correct answers or the question was ambiguous, then it is excluded from scoring and the resident can ignore it entirely.

Figure 2a. Item Performance Report (Blueprint Category)

	Acute Care and Diagnosis	Chronic Care Management	Emergent and Urgent Care	Preventive Care	Foundations of Care
1000		009 183	013		
950			079		
850	050 089		127		
800	026				
750	174	073 196		002 063	
700	170 186	094 112	041 160	119 187	
650	015 020	046 121 185	123	088 113 182	106
600	124 126 152	037 077 165	098 154	034 095	
550	018 130 132	025 085	004		137
500	038	017 042 069 125	076 184		
450	005 012 048 067 093 114 178 193	051 134	030	039 055 083 153	033
400	087 091 118	131 198	102 151 199	049 169	
350	036 078 082 136 162 180 197 200	179	058 070 171	021	
300	001 003 031 056 060 084 111 147 158 164	075 107 128 144 156 172	035P	100 110	
250	024 072 096 101 140 155	014 057 071 190	010 019 027 148 188 192	129 177	
200	008 040 104 189	011 054 097 150	092		043 081
150	022 065 143 149	032 139	138	028	
100	062 080	175	023 047 066 142		117 173
50		061 159	086	090	016
0	029 044 053 099 109 116 122 145	105 146 161 168 181	052 108 115 120 133 166 176 195	007 045 068 074 103 135 141 157	059 194

Figure 2b. Item Performance Report (Body System Category)

	Cardiovascular	Injuries/Musculoskeletal	Respiratory	Psychiatric/Behavioral	Sexual and Reproductive
1000		009	013	183	
950					
850	050	089			
800					
750				073 196	174
700		160	186	112	094
650		123	185		015 046
600		098		077	
550		130	018 132	085	
500	125	017 042 184			
450	005 134	178 193			
400		087 102			131
350	162	058 082 136	171		179
300	128		056 111	156	107 147
250	019 024 188 192	027	101	014 071	057
200	011 092	189			104
150			139		022
100	023 066 142			175	
50	159			061	
0	146 176	099 108 168	122	052 133 166	

Suggestions for Reviewing Scores with Residents

1. Primarily attention should be paid to a resident's overall scaled score. It is based on the resident's responses to all the scored questions and is the best estimate of how the resident would perform on the FMCE at that point in time. Identifying residents who need more assistance early on in residency allows for more time to get them back on track. The Bayesian Score Predictor (BSP, Section V) can also help to identify them.
2. Because a resident's scaled score is reported on the same scale across all years of residency as well as on the certification examination, using these scaled scores to monitor a resident's progress throughout training will provide an effective way for residents to get timely feedback. Resident performance should be compared to the minimum passing score (MPS) for the FMCE. Presently, the MPS is 380. Typically, ITE score can be expected to jump approximately 40-60 points from PGY1 to PGY2 and another 25-35 points from PGY2 to PGY3. Third year residents that score a 440 or better generally have a very good probability of passing the certification examination.[1-2]
3. To identify the areas of study that will produce the largest gains for them, residents should identify the content areas that have the most questions. This information is available on the score report under *Exam Blueprint Percentage* column. Of course, this should not be construed as meaning that the resident should neglect the other areas of study.
4. After ABFM has released the ITE questions to the residency programs, it is useful for the resident to go through the ITE question-by-question and
    a. remember what answer they chose and why,
    b. look up the correct answer in the critique,
    c. if the resident's answer was wrong, identify what was the *disconnect* between their rationale and the rationale in the critique,
    d. review the disconnects to see if any common themes emerge. **Note:** The disconnect could be a deficit in knowledge or it could be related to a test-taking style issue, such as not reading the question carefully, reading too much into the question, coming to a conclusion too quickly, answering a similar question, but not exactly the question asked, etc.
    e. Create a study plan based upon disconnects.
5. Do not rely heavily on sub scores. ITE sub scores are not sufficiently precise to reliably confirm knowledge deficiencies within a content area. This may be readily understood by examining the very large standard error associated with many of the content categories. However, these scores may be useful in the context of generating (but not confirming) hypotheses about an individual's strengths and weaknesses of knowledge.

[1] O'Neill TR, Li Z, Peabody MR, Lybarger M, Royal KD, Puffer JC. The Predictive Validity of ABFM's In-Training Examination. *Fam Med.* 2015;47(5):349-356.

[2] O'Neill TR, Peabody MR, Song H. The predictive validity of the National Board of Osteopathic Medical Examiners' COMLEX-USA examinations with regard to outcomes on American Board of Family Medicine examinations. *Academic Medicine*. 2016;91(11):1568-1575.

SECTION II

Program-level Reports

Overall Scaled Score Distribution Graphs
In Figure 3, the 2024 ITE reported scaled scores for the entire population of residents enrolled in ACGME-accredited programs are represented with a panel specific to each training year and a panel representing the combined population across all training years. The reported scaled scores increase in increments of 10. Scores lower than 200 are reported as 200 and scores greater than 800 are reported as 800. Generally, the panels show a normal, symmetrical distribution (with PGY4 as exception due to small sample size). ACGME-accredited family medicine residencies are typically a 3-year curriculum. The PGY4 chart in Figure 3 represents the reported scaled score of those PGY4 residents in a Length of Training (LoT) pilot program. It does not represent those residents who are in a dual-specialty program (Family Medicine & Psychiatry, Family Medicine & Internal Medicine, etc.)

Mean Scaled Score Graph
Each program's *Mean Scaled Score Graph* (Figure 4) shows the mean scaled score for the population and then mean scaled score for the program within each training year as well as for all training years combined.

Eight Year Trend Chart and Data
The 8-year trend summary data from Figure 5 is visually depicted in Figure 6. Each data point in the table (Figure 5) represents the height of a bar on the graph, making trends in the performance of your program both across PGY cohorts and overtime within cohorts more obvious. Comparisons to national averages are also presented. The data table presented happens to be the Microsoft Excel version, but a comma-delimited version is also available, in case you have access to or a preference for other spreadsheet software.

The graph (Figure 6) shows a program's mean scaled scores over the past eight years. Please note that programs new to the ITE may not yet have ITE scores for all years, but the national averages are still presented. Within each calendar year, your program's average score is shown adjacent to the national average for that group. Within calendar year, the mean scores for both your program and the nation are presented broken out by each training year (PGY1, PGY2, PGY3, and PGY4) and aggregated across PGY1 to PGY4. Please note a typical family medicine residency is 3 years long; however, there are several residencies that are participating in the Length of Training (LoT) pilot which provides 4 years of family medicine training. The PGY4 cohort represents only these residents. Although the graph is aesthetically busy, it provides a great deal of information.

Figure 3. Scaled Score Distribution

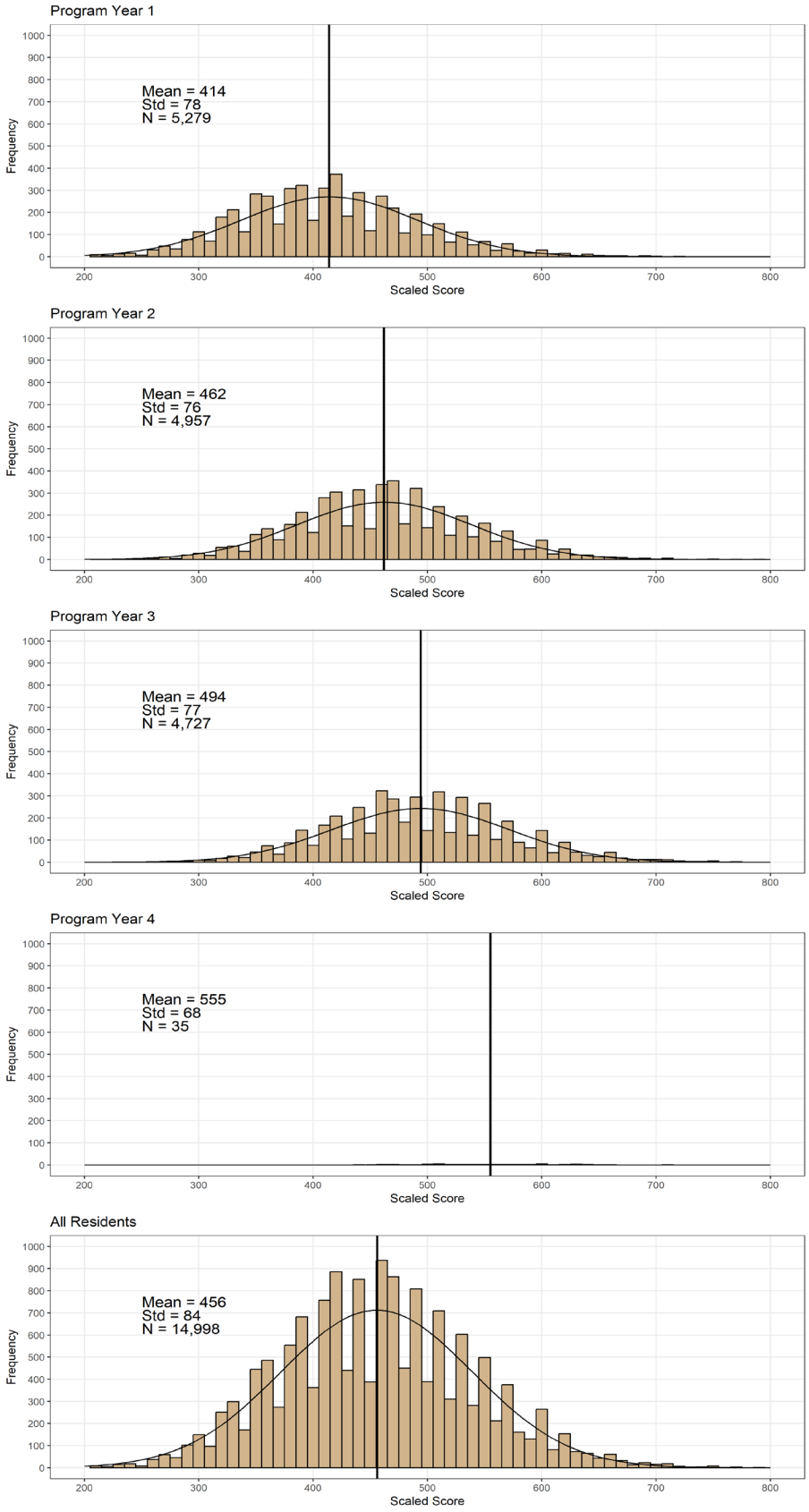


Figure 4. Example of a Program-Level Report

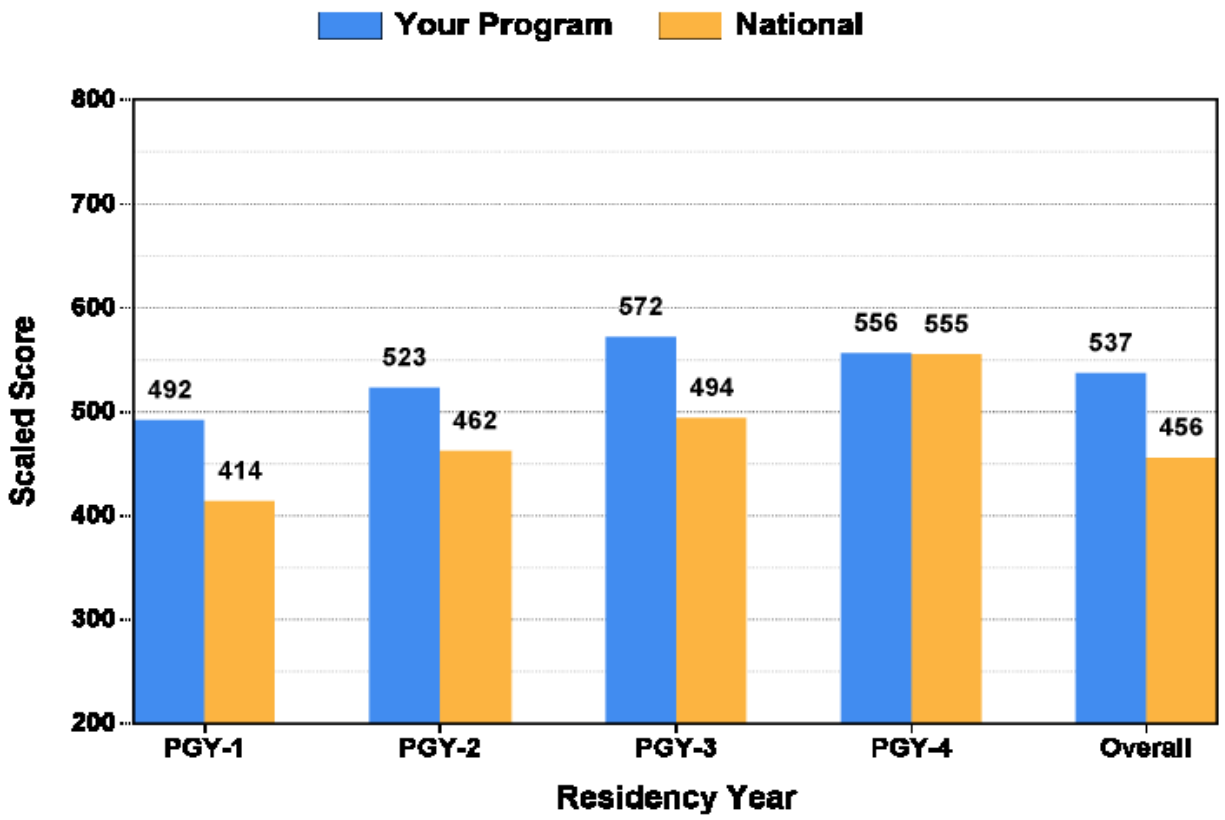
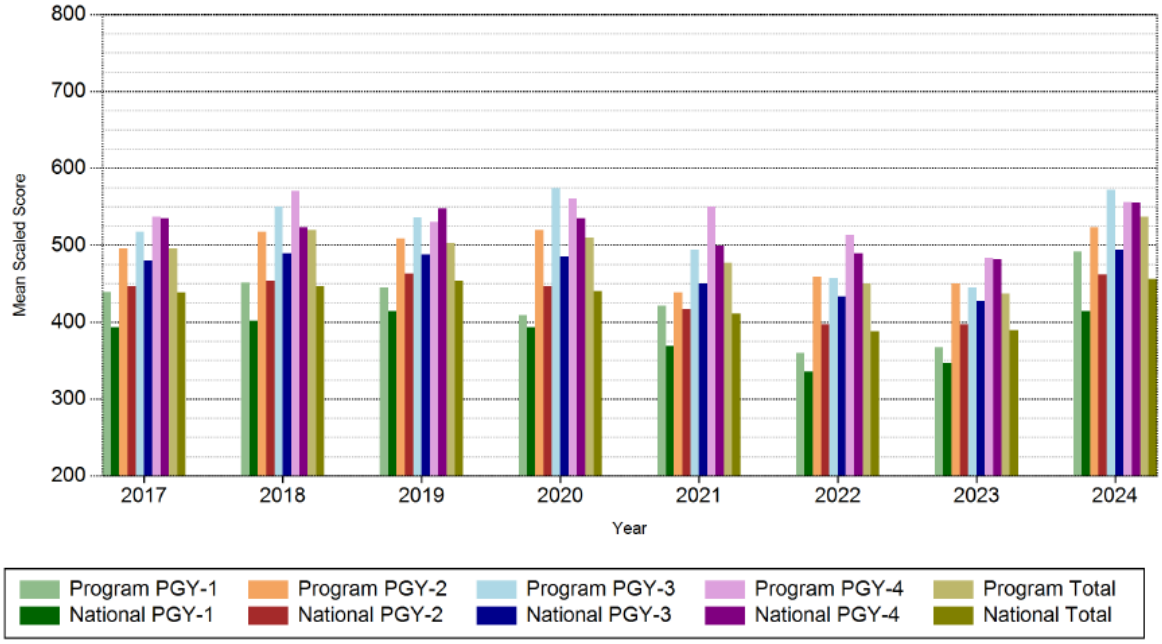


Figure 5. Data Table for 8 Year Trend Chart

	Exam Year	PGY-1	PGY-2	PGY-3	PGY-4	Total
National	2017	393	447	480	535	438
Program	2017	439	496	517	537	496
National	2018	402	454	489	523	447
Program	2018	451	517	550	571	520
National	2019	414	463	488	548	454
Program	2019	445	509	536	530	503
National	2020	393	447	485	535	440
Program	2020	409	520	574	560	510
National	2021	369	417	450	499	411
Program	2021	421	438	494	550	477
National	2022	336	397	433	489	388
Program	2022	360	459	457	513	450
National	2023	347	397	427	482	389
Program	2023	367	450	445	483	437
National	2024	414	462	494	555	456
Program	2024	492	523	572	556	537

Figure 6. ITE 8-Year Trend Chart



The ITE was scored using procedures designed to connect the ITE results to the scale of the certification examination.

SECTION III

Individual-Level Score Report Data for All Residents

Introduction
In Figures 7 and 8, examples of two data tables are presented to illustrate the values seen on each residents' detailed score report (see Figure 1). These data are provided in both Microsoft Excel and comma-delimited format. The example below uses the Excel format, which comes as two worksheets within a single workbook. The first worksheet (Figure 7) presents the data table. The second worksheet (Figure 8) shows the descriptive labels for the columns in the data table. In the comma-delimited version of this file, only the data is included. To get the labels, you must use the second excel worksheet in the workbook. The numbers seen in columns of the data table correspond to the codes in the label table. For example, columns in the data table beginning with ID101 provide values describing the "Total" set of questions on the examination, whereas columns beginning with ID201 are in regard to "Acute Care and Diagnosis" questions.

Row Descriptions
As seen in Figure 7, you can compare your program's performance (row 9) to the national averages for each year of residency (rows 6-8). Row 9 shows your program's ID along with data that applies to the program as a whole. Row 10 describes your first-year residents' performance followed by rows with the results for named first year residents. This pattern is repeated for the second, third, and sometimes fourth year residents in subsequent rows.

Column Descriptions
Column A shows whether a particular set of rows provides data for national values or for your program (designated by your Program ID#). Column B shows the relevant training year that is being described in each row. Column C provides residents' names. Column D provides the resident's ABFM ID number.

Columns E through I (ID101) represent performance on the entire test. Columns J through BG represent subtest performance. The same variables that are available for the entire test (ID101) is also available for blueprint subtests (ID201 to ID205), as well as for the body system categories (ID301 to ID 305). Descriptions of what these ID###s represent can be found in Figure 8.

Each consecutive set of 5 columns to the right show the number of questions in the indicated category (as in column E), the raw number of questions correct in that category (as in F), the percent correct within the category (as in G), and the scaled score (as in H) which is based upon the number correct and the difficulty of the questions in that content area, and the standard error of the scaled score (as in I).

Figure 7. Individual-Level Score Report Data Table for All Residents

	A	B	C	D	E	F	G	H	I	J
1	2024 ITE Score Data									
2	ITE Year: 2024									
3										
4	ACGMEID	Training Year	Full Name	ABFMID	ID101 Item #	ID101 Raw Score	ID101 Percent	ID101 Scaled Score	ID101 SEM	ID201 Item #
5	National				194	119	61.1	456	37	68
6		1			194	112	57.8	414	37	68
7		2			194	120	61.7	462	37	68
8		3			194	125	64.3	494	38	68
9	5555555555				194	113	58.4	425	37	68
10		1			194	101	52.1	348	36	68
11			Doogie Howser, MD	555001	194	106	54.6	380	36	68
12			Nicholas Riviera, MD	555002	194	95	49.0	310	36	68
13			Julius Hibbert, MD	555003	194	96	49.5	320	36	68
14			Gregory House, MD	555005	194	107	55.2	380	36	68
15			John McIntyre, DO	555006	194	117	60.3	450	37	68
16		2			194	147	75.8	650	41	68
17			B.J. Hunnicutt, MD	555008	194	101	52.1	350	36	68
18			John Zoidbert, MD	555009	194	118	60.8	450	37	68
19			Leonard McCoy, MD	555010	194	113	58.2	420	36	68
20			John Watson, MD	555011	194	106	54.6	380	36	68
21			Marcus Welby, MD	555012	194	124	63.7	487	38	68
22		3			194	109	56.2	390	36	68
23			Henry Blake, DO	555015	194	141	72.7	600	40	68
24			Frank Burns, MD	555016	194	121	62.4	470	37	68

Figure 8. Labels for the Data Table

	A	B
1		
2	Item Name	Description
3	ID101	Total Questions
4	ID201	Acute Care and Diagnosis
5	ID202	Chronic Care Management
6	ID203	Emergent and Urgent Care
7	ID204	Preventive Care
8	ID205	Foundations of Care
9	ID301	Cardiovascular
10	ID302	Injuries/Musculoskeletal
11	ID303	Respiratory
12	ID304	Psychiatric/Behavioral
13	ID305	Sexual and Reproductive

SECTION IV

INSTRUCTIONS FOR SCORING BY HAND

Introduction

Scoring the ITE by hand requires that you have answered ALL the questions on the examination. Because there is noticeable variability in the difficulty of the questions, one must consider BOTH how many questions were answered correctly and how difficult the questions were. The raw to scaled score conversion table (Table 3) applies only to ENTIRE test. There are no conversion tables available to the end user that are specific to subsets of questions selected by the end user.

Key Changes and Questions Removed from Scoring

A psychometric evaluation of each question is performed following the initial administration of each exam form. Whenever a question is not functioning as expected, the question, critique, references, and answers are carefully reviewed by a content expert prior to the final scoring of the examination. For the 2024 ITE, no answers had to be re-keyed, one question was deleted for psychometric reason, and five questions were deleted for content reason. After removing these questions, the final number of questions scored was 194.

Questions Deleted	Reason for Removal
Question 6	Content Delete
Question 64	Content Delete
Question 163	Content Delete
Question 167	Content Delete
Question 191	Content Delete
Question 35	Psychometric Delete

Scoring by Hand

To score the examination by hand is a rather slow process; however, it is very easy and quite intuitive.

Step 1. Take the test and record the answers.

Step 2. Using the Critique Book as the answer key, mark all the questions as RIGHT or WRONG. Unanswered questions are wrong.

Step 3. Count how many questions were answered correctly, excluding any question removed from scoring (content or psychometric deletes).

Step 4. Look up in Table 3 (Raw Score to Scaled Score Conversion table), the number of questions answered correctly in the raw score column. The corresponding number in the Scaled Score column is your scaled score.

Table 3. Raw Score to Scaled Score Conversion

Raw Score	Scaled Score	Raw Score	Scaled Score	Raw Score	Scaled Score
0-77	200	111-112	410	143-144	620
78-79	210	113-114	420	145	630
80	220	115	430	146	640
81-82	230	116-117	440	147	650
83-84	240	118	450	148-149	660
85	250	119-120	460	150	670
86-87	260	121-122	470	151	680
88-89	270	123	480	152	690
90	280	124-125	490	153	700
91-92	290	126	500	154-155	710
93-94	300	127-128	510	156	720
95	310	129	520	157	730
96-97	320	130-131	530	158	740
98-99	330	132	540	159	750
100	340	133-134	550	160	760
101-102	350	135	560	161	770
103-104	360	136-137	570	162	780
105	370	138	580	163	790
106-107	380	139	590	164-194	800
108-109	390	140-141	600		
110	400	142	610		

Table 4 is included so that people attempting to administer this exam via pencil and paper can label the questions by content category for review purposes.

Table 4. Questions by Blueprint Category

Acute Care and Diagnosis	1, 3, 5, 8, 12, 15, 18, 20, 22, 24, 26, 29, 31, 36, 38, 40, 44, 48, 50, 53, 56, 60, 62, 65, 67, 72, 78, 80, 82, 84, 87, 89, 91, 93, 96, 99, 101, 104, 109, 111, 114, 116, 118, 122, 124, 126, 130, 132, 136, 140, 143, 145, 147, 149, 152, 155, 158, 162, 164, 167, 170, 174, 178, 180, 186, 189, 191, 193, 197, 200
Chronic Care Management	6, 9, 11, 14, 17, 25, 32, 37, 42, 46, 51, 54, 57, 61, 64, 69, 71, 73, 75, 77, 85, 94, 97, 105, 107, 112, 121, 125, 128, 131, 134, 139, 144, 146, 150, 156, 159, 161, 163, 165, 168, 172, 175, 179, 181, 183, 185, 190, 196, 198
Emergent and Urgent Care	4, 10, 13, 19, 23, 27, 30, 35, 41, 47, 52, 58, 66, 70, 76, 79, 86, 92, 98, 102, 108, 115, 120, 123, 127, 133, 138, 142, 148, 151, 154, 160, 166, 171, 176, 184, 188, 192, 195, 199
Preventive Care	2, 7, 21, 28, 34, 39, 45, 49, 55, 63, 68, 74, 83, 88, 90, 95, 100, 103, 110, 113, 119, 129, 135, 141, 153, 157, 169, 177, 182, 187
Foundations of Care	16, 33, 43, 59, 81, 106, 117, 137, 173, 194

This list is specific to the 2024 ITE. It cannot be used with tests from other years.

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SECTION V

Bayesian Score Predictor

Overview

The ABFM created the Bayesian Score Predictor (BSP) to provide program directors and residents with information to help interpret what a resident's ITE score implies about their future performance on the FMCE. The ITE score can be interpreted directly as our best estimate of how the resident would perform on the FMCE at that point in time. Of course, the score is only an imprecise estimate. Given that the standard error of measurement (SEM) for the ITE is roughly 38 points, the observed ITE score has a 68% chance of being within 38 points of the resident's true ability and a 95% of being within 76 points of the resident's true ability. The imprecision described by the SEM is just an articulation of the bounds of the random sampling process. It does not consider any additional factors (fatigue, illness, extraordinary distractions, etc.) that might make the observed ITE score a less accurate estimate of the resident's true ability. With regard to predicting whether an individual will pass the FMCE, ITE scores over 440 are very re-assuring, assuming that the ITE was taken under test-like conditions. By test-like conditions, we mean that (1) the examinee has not seen the ITE questions prior to taking the test, (2) the examinee takes the test within the allotted time constraints, and (3) the examinee does not use external resources, including people, to look up the answer to the questions. For additional information regarding the predictive value of the ITE and the COMLEX-USA for predicting performance on the FMCE, please see the citations in the footnotes[3,4].

Although directly interpreting the ITE score has the advantage of providing an estimate of how the examinee would perform on the FMCE, it does not provide a probability of passing. For example, if 380 is passing for the FMCE and a PGY3 resident scored 380 on the ITE, what is the probability of them passing the FMCE? It is not 100%! It is more like 85%[5]. Also, consider the case of a second-year resident who gets an ITE score of 350. Is this resident on track to pass the FMCE? The ABFM Bayesian Score Predictor (https://rtm.theabfm.org/bayesian/predictor) can help answer these questions.

Using the Bayesian Score Predictor

The Bayesian Score Predictor (BSP) is intended to provide family medicine residents and program directors with an estimate of the resident's probability of passing the ABFM certification examination when they take it at the end of PGY-3. The program's user interface (Figure 9a) has six boxes. The first box contains either a range of COMLEX-USA® Level 2 CE scores or a range of USMLE Step 2 CK scores with the expected distribution of examinees for those scores. You

[3] O'Neill TR, Li Z, Peabody MR, Lybarger M, Royal KD, Puffer JC. The Predictive Validity of ABFM's In-Training Examination. *Fam Med*. 2015;47(5):349-356.

[4] O'Neill TR, Peabody MR, Song, H. The Predictive Validity of the National Board of Osteopathic Medical Examiners' COMLEX-USA Examinations With Regard to Outcomes on American Board of Family Medicine Examinations. *Acad Med*. 2016; 91(1): 568–1575.

[5] Please note that in Section I, we indicated that someone who just barely earned a score of 380 would theoretically have a 50% chance of passing if they took the examination at that point in time. Here we are discussing, everyone who earned a 380 (just barely to almost a 390) and assuming that they still have an additional 5 months in their residency program before taking the certification examination.

can choose which one you wish to enter. The next three boxes contain the useful range of scores on the ITE, one box for each year of residency. The next box, the fifth one, represents the useful range of scores on the FMCE and the distribution of examinees associated with these scores. The sixth and final box indicates the probably of passing the FMCE. To really understand what the BSP is doing, it helps to play with it for a while. Do not worry about breaking it. It is a web-based application, so it is difficult to break.

In its initial state, the boxes are populated with a useful distribution of licensing examination, ITE, and FMCE scores (Figure 9a). However, if the user clicks on a score in any one of these boxes (Figure 9b) the distribution changes from a distribution of scores to the selected observed score. This will in turn cascade forward to revise the distributions to reflect the expected ITE scores in later years of residency and then to the expected distribution of FMCE scores. These revisions represent the probabilities that are expected based upon only those people who scored the selected score. Please note that when you click on a score to indicate that it is the score you earned; the observed score has a 100% probability of having happened because you said that it did. This directs the BSP to consider only the distributions that would result from the population of people who earned that observed score.

As an example, consider the case of a resident who scored 270 in PGY-2 (Figure 9b). Here, the user would click on <270> in the PGY2 box. This will (1) change the PGY2 distribution to 100% for the score of 270 and (2) change the expected score distributions for PGY3 and for the FMCE to reflect the percentages expected for a resident who scored a 270 in PGY2. You can see that this also updates the resident's probability of passing.

Furthermore, if the user later clicks on a different PGY2 score, say 410 (Figure 9c), the program will change the "observed score" to 410 and update the distributions and probabilities accordingly. A second click on 410 restores the original default distribution for PGY2. It is important to note that if a score is selected for PGY1 and then for PGY2, the PGY2 score overrides the PGY1 score because it occurred at a point in time closer to when the FMCE pass-fail decision is made.

As another example, let us say that a resident in PGY1 earned a 340 on the ITE. Clicking on <340> in the PGY1 Score column will show the expected future score distributions. This will also cause the other four boxes re-populate and the resident's expected pathway can be understood as the mean score found in PGY2, PGY3, and on the FMCE. The acceptability of that expected pathway is really a question of how satisfied the resident and program director are with the probability of passing the FMCE.

It is also important to note that probabilities cascade backward as well as forward, showing the distribution of scores that likely generated the observed score. This is not helpful for making predictions about passing, but it might be helpful in quantifying the degree of unexpectedness when something strange happens, such as a low scorer in PGY 1 and 2 scoring unexpectedly high in PGY3.

If you have questions regarding the Bayesian Score Predictor, please contact the ABFM psychometrics department.

Figure 9a. Bayesian Score Predictor in its Initial State

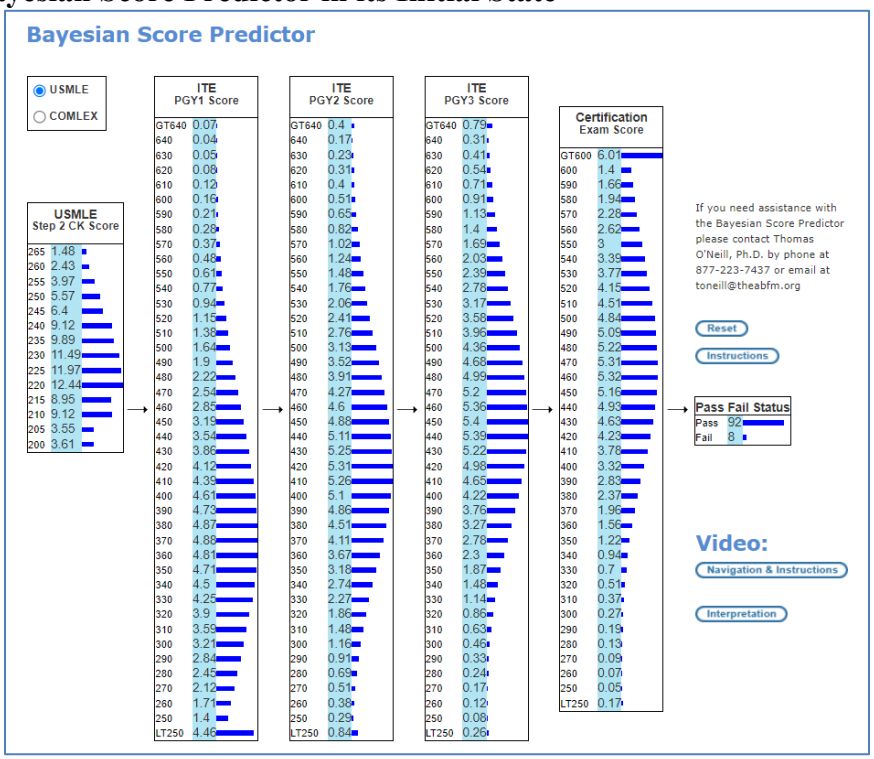


Figure 9b. Bayesian Score Predictor with PGY2 = 270

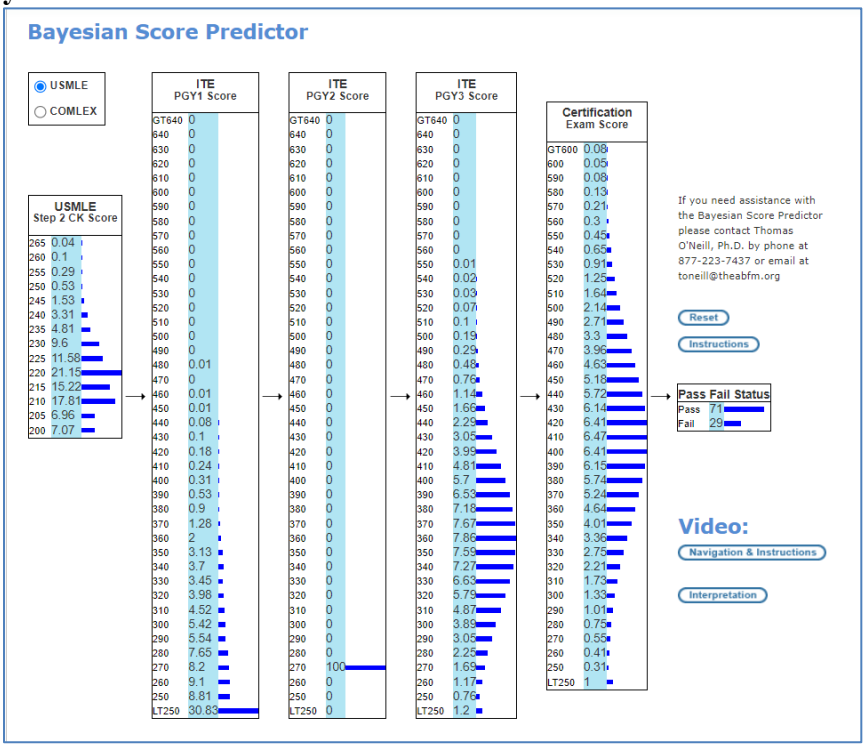
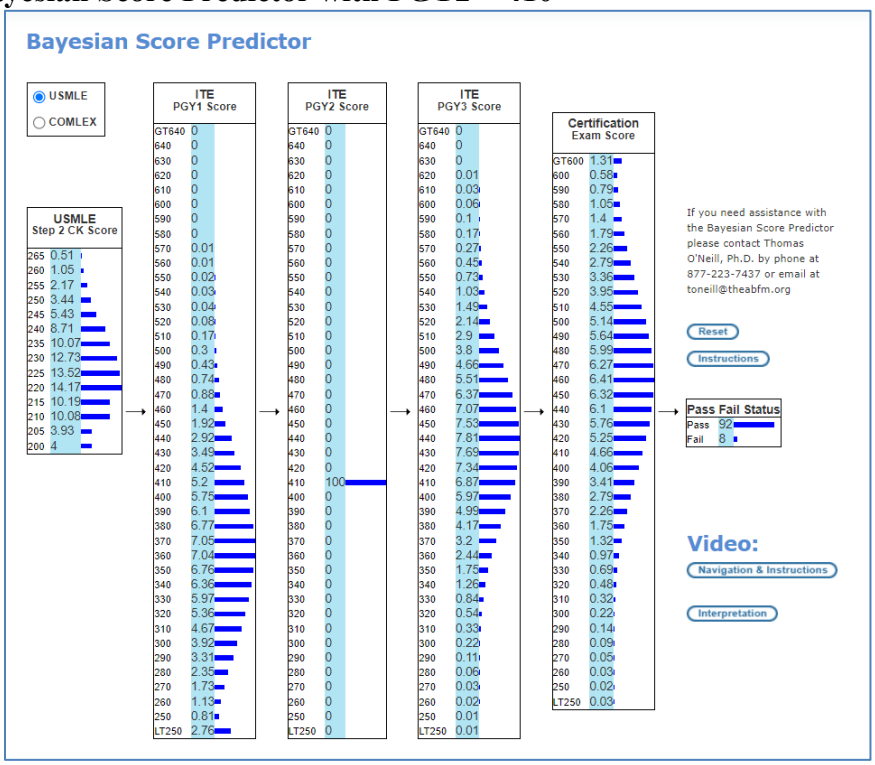


Figure 9c. Bayesian Score Predictor with PGY2 = 410



SECTION VI

Psychometric Summary

Purpose

The ABFM offers residents enrolled in ACGME-accredited residency programs the opportunity to take the ITE. The purposes of the ABFM's ITE are: (1) to provide each resident with a low-cost, low-stakes opportunity to become familiar with the general format and item writing style that will be used on the FMCE and (2) to provide each resident and his or her program director with an opportunity to assess how well the resident is progressing toward eventually passing the FMCE. As the ITE is designed as a low-stakes examination, the ABFM advises program directors that the results should not be used to make important decisions related to the promotion or advancement of residents taking the exam. To achieve the purposes stated above, it is important that: (1) the ITE content be similar to that of the FMCE, (2) the ITE results be a good approximation of how a resident would have performed on the FMCE at that point in time, and (3) the ITE be predictive of an examinee's future results on the FMCE.

It is important to note that ABFM has conducted research to connect ITE scores with the ABFM's certification exam results so that reasonable predictions can be made. However, the research is silent about making predictions about how well the ITE predicts performance on the American Osteopathic Board of Family Physicians (AOBFP) certification exam. Similarly, our research is also silent on how well the American College of Osteopathic Family Physician's (ACOFP) In-Service Exam (ISE) predicts performance on ABFM certification examination. ABFM just does not have the data for those analyses.

Criterion-related Validity

The ABFM's ITE was designed to have a high degree of concurrent and predictive validity, both of which are forms of criterion-related validity. The criterion in both is performance on the FMCE. The concurrent validity claim for the ABFM's ITE is that it is intended to produce scores that would be predictions of how an examinee would perform on the FMCE if he or she had taken it instead of the ITE at that point in time. To achieve this, each form of the ITE is built to the same specifications as the FMCE and the ITE scores are equated onto the FMCE scale using the dichotomous Rasch model and a common-item equating design. Concurrent validity with the FMCE is partially established through the regular quality checks that ensure the ITE is developed with the correct content specifications and that the equating was successful. Because the results of these quality control checks have always been quite good, the ABFM has never conducted an experiment in which both tests were administered to examinees on consecutive days; however, the ABFM does administer the FMCE a few months after PGY3 residents take the ITE, which is how the ABFM usually establishes the predictive validity of the ITE[6].

Test Specifications

The ITE contains 200 multiple-choice questions and is built to mirror the same content domain proportions as the FMCE. The process used to develop the content specifications for the FMCE is

[6] O'Neill TR, Li Z, Peabody MR, Lybarger M, Royal KD, Puffer JC. The Predictive Validity of ABFM's In-Training Examination. *Fam Med*. 2015;47(5):349-356.

described in greater detail by Norris, Rovinelli, Puffer, Rinaldo, and Price[7]. Since 2013, it has been administered only in a computer-administered format. Each year, there is a different form of the ITE with no questions in common from previous forms because after each administration, the questions on that test are made available to residents and diplomates for educational purposes. These questions are then no longer secure and would probably be known by the examinees if they were reused.

Equating the ITE to the FMCE

To equate the ITE across administrations and to make the ITE score represent the examinee's predicted performance on the FMCE, the ABFM includes a small number (approximately 40) of ITE questions as unscored pretest questions on the FMCE. These questions are then calibrated onto the FMCE scale. Displacement[8,9] indices are used to check that the relative question difficulty calibrations derived from the FMCE examinee population still hold for the ITE examinee population. In cases where the displacement is more than 0.6 logits, the question is no longer considered an anchor and the ITE data is used to recalibrate it. In this way, these ITE questions and their associated FMCE calibrations are used to connect each administration of the ITE to the continuously maintained Family Medicine Certification - Scale (FMC-Scale).

Scoring

Prior to scoring, all 200 questions are subjected to an item analysis to flag potentially problematic test questions. Questions that are determined to be inadequate for content reasons are removed from scoring. Questions that are determined to psychometrically function in a manner that degrades the measurement quality of the ITE are also excluded. These questions are listed in Section IV.

The remaining ITE questions are scored as right or wrong and these scored responses are transformed into a logit metric using the dichotomous Rasch[10] model. Next, the logit values are converted into scaled scores that range from 200-800. Scores below 200 are set to 200 and scores above 800 are set to 800. The Rasch model, in conjunction with a common item equating design, is also used to equate examinations across test forms and years of administration. The use of a common scale with a passing standard that is held constant for useful periods of time has the advantage of providing a more stable target for making predictions related to whether a candidate will pass or fail.

2024 Administration

The ABFM's ITE is administered to nearly all family medicine residents in ACGME accredited programs. In 2024, there were 14,998 residents from 741 residency programs that took the ITE. The number of residents in each year of residency was evenly distributed.

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[7] Norris TE, Rovinelli RJ, Puffer JC, Rinaldo J, Price DW. From Specialty-Based to Practice-Based: A New Blueprint for the American Board of Family Medicine Cognitive Examination. *Jounal of the American Board of Family Practice*. 2005;18(6):546-554.

[8] Linacre, J. M. (2013). *Winsteps® Rasch measurement computer program User's Guide*. Beaverton, Oregon: Winsteps.com.

[9] O'Neill, TR, Peabody, MR, Tan RJB, Du, Y. How much item drift is too much? *Rasch Measurement Transactions*, 2013, 27:3 p. 1423-4.

[10] Rasch G. *Probabilistic models for some intelligence and attainment tests*. Copenhagen, Denmark.: Danish Institute for Educational Research; 1960.

Reliability and Person/Item Fit

When a test is administered, it is desirable to have some assurance that comparable results would be obtained if the same people were tested again under similar conditions. This reproducibility of test scores is referred to as reliability. It is important to note that reliability is not a property of the test, but rather a property of the data set. It is influenced by both the sample of candidates and the test. When using the Rasch model, it can be thought of as how well the questions spread out the people (person reliability, the conventional usage) and how well the people spread out the questions (item reliability, sometimes considered by content developers, but not by examination end-users). Given the data from the 2024 ITE, 1.97 strata of candidates (Table 5) were discernible, which is comparable to a reliability of 0.80. With regard to the questions, 56.01 strata (Table 6) were discernible, which is comparable to a reliability of 1.00.

Model fit refers to how well the data conform to the requirements of measurement. This is considered both for examinees and for questions. A summary of fit can be found in Table 5 for candidates and Table 6 for questions. Four statistics are considered: Information Weighted Mean Square (INFIT MNSQ); Information Weighted Mean Square Standardized (INFIT ZSTD); Unweighted Mean Square (OUTFIT MNSQ); and Unweighted Mean Square Standardized (OUTFIT ZSTD).

The mean square statistics have an expected value of 1.0, but can range from 0 to infinity. They can be thought of as an indicator of the magnitude of the disturbances in the data. The standardized mean square statistics have a mean of 0 and a standard deviation of 1. They can be thought of as indicators of whether the disturbance could be attributed to chance. The weighted statistics are more sensitive to disturbances near the person's ability (or the question's difficulty). The unweighted statistics are more sensitive to outlying disturbances such as high ability people missing an easy question or a low ability people making more lucky guesses than expected.

The magnitude of disturbances among the questions did not rise to a level to warrant concern; thus, the issue regarding whether the disturbance is attributable to chance is irrelevant. There were some questions that exhibited larger than preferred levels of misfit; however, those questions also had high levels of displacement, but did not reach the threshold at which they would be unanchored (0.6 logits).

Table 5. Summary of 2024 ITE Candidates (N = 14, 998)

	Ability[1]	SE	INFIT MNSQ	INFIT ZSTD	OUTFIT MNSQ	OUTFIT ZSTD
Mean	461	37	1.00	0.07	1.00	0.01
S.D.	87	1	0.08	1.07	0.16	0.99
Max	805	48	1.53	5.10	2.53	6.90
Min	1	36	0.72	-3.71	0.58	-2.64

Note: Separation, the ratio of the precision of the candidate ability estimates to the adjusted spread of ability among candidates, was 1.97. This is a ratio measure of separation in Root Mean Square Error units. The separation expressed as a reliability coefficient (comparable to a KR20 or Cronbach's alpha) was 0.80.

[1]This is based on the true ability, no ceiling/flooring limits

Table 6. Summary of 2024 ITE Questions (N = 194)

	Difficulty	SE	INFIT MNSQ	INFIT ZSTD	OUTFIT MNSQ	OUTFIT ZSTD
Mean	327	5	1.01	0.51	1.00	0.14
S.D.	304	2	0.07	5.00	0.09	5.50
Max	1348	25	1.53	9.90	1.74	9.90
Min	-711	4	0.72	-9.90	0.71	-9.90

[*]Separation, the ratio of the precision of the question difficulty estimates to the adjusted spread of difficulty among questions, was 56.01. This is a ratio measure of separation in Root Mean Square Error units. The separation expressed as a reliability coefficient (comparable to a KR20 or Cronbach's alpha) was 1.00.

END OF REPORT